

Input Shaping for Robust Vibration Suppression in PPP Cartesian Robots with Variable Z-Axis Dynamics for Automated Optical Inspection [†]

Demian Ecurra, Luis M. Prieto, Luis E. Vázquez, Shu-Hao Liang and Marnel Altius

¹ Graduate Institute of Intelligent Manufacturing Tech., National Taiwan University of Science and Technology, Taipei City 106335, Taiwan; shuhaoliang@mail.ntust.edu.tw

² Department of Mechanical Engineering, National Taiwan University of Science and Technology, Taipei City 106335, Taiwan

* Correspondence: eliazartev84@gmail.com

[†] Presented at the 6th International Conference on Electronic Communications, Internet of Things and Big Data, April 24–26 2026, Tamkang University, New Taipei, Taiwan

Abstract

High-speed point-to-point motion in PPP Cartesian robots for Automated Optical Inspection (AOI) of printed circuit boards induces residual vibrations that force the system to wait until the camera head settles before image acquisition can proceed, directly limiting inspection throughput. While input shaping is a well-established feedforward strategy for vibration suppression in flexible systems, its effectiveness in reconfigurable Cartesian platforms is complicated by workspace-dependent natural frequency variation across both Z-axis extension and Y-axis position. This work presents the design, simulation, and experimental validation of a Zero Vibration and Derivative (ZVD) input shaper implemented on an ESP32 microcontroller for a PPP Cartesian AOI robot. The dominant flexible mode of the X-axis belt-drive was identified experimentally from post-stop accelerometer ring-down data across nine workspace configurations using FFT-based frequency identification and logarithmic decrement damping estimation, yielding a mean natural frequency of 15.34 ± 1.9 Hz and damping ratio of 0.032 ± 0.007 . A per-Y-position ZVD shaper lookup table was deployed on the robot and validated across seven configurations spanning three Z positions and three Y positions. Experimental results demonstrate RMS residual vibration reductions ranging from 5.3% to 70.7%, with a mean of 43.7% across all configurations and a maximum peak reduction of 79.1%. A simulation-based robustness sweep confirms that configurations falling within a frequency ratio of 0.7–1.4 relative to the design frequency achieve consistent suppression, providing a practical design guideline for workspace-varying systems.

Keywords: Input Shaping; Residual Vibration Suppression; Cartesian Robots; ZVD Shaper; Time-Varying Dynamics

This research was supported by the Intelligent Manufacturing Innovation Center (NTUST) within the Featured Areas Research Center Program and the Higher Education Sprout Project Framework of the Ministry of Education, Taiwan.

Copyright: © 2026 by the authors. Submitted for possible open access publication under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

1. Introduction

Automated Optical Inspection (AOI) of printed circuit boards (PCBs) demands both high positioning accuracy and high throughput. In Cartesian robot platforms, these requirements are in direct tension: increasing motion speed reduces cycle time but amplifies residual vibrations at the end of each point-to-point move, forcing the system to wait until the camera head settles before image acquisition can proceed. This settling time represents

dead time in the inspection cycle and is the primary bottleneck for throughput in high-mix, low-volume (HMLV) production environments.

The dynamic behavior of belt-driven Cartesian robots is governed by structural flexibility and transmission compliance. The belt-drive mechanism introduces a resonant flexible mode whose natural frequency depends on belt stiffness, cart mass, and motor-side inertia, as described by the two-mass model [1,2]. Rapid trapezoidal motion commands excite this mode, producing a decaying oscillation after each stop that the accelerometer at the camera head measures directly as residual acceleration. In PPP configurations with a variable-length Z-axis, the structural compliance of the vertical arm changes with Z-extension, shifting the natural frequency and complicating fixed-parameter vibration suppression.

Among feedforward vibration mitigation strategies, input shaping has demonstrated consistent effectiveness across a wide range of flexible systems [3]. The method works by convolving the motion command with a sequence of impulses designed to cancel the system's residual oscillation at the end of the move — requiring no additional sensors or feedback hardware beyond what is already present for motion control [4]. The Zero Vibration and Derivative (ZVD) shaper extends this concept by additionally constraining the derivative of residual vibration with respect to frequency, yielding robustness to modeling errors that makes it well-suited to systems with uncertain or slowly varying natural frequencies [3]. Input shaping has been validated experimentally on Cartesian robots [5] and more recently on industrial robot arms achieving residual vibration reductions exceeding 85%.

This paper presents the design, simulation, and experimental validation of a ZVD input shaper for a PPP Cartesian AOI robot. The dominant flexible mode of the X-axis belt-drive is identified experimentally from post-stop ring-down measurements at multiple workspace configurations. A fixed-parameter ZVD shaper is designed from these measurements and implemented on the robot's microcontroller. Simulation using a base-excitation second-order model demonstrates near-complete cancellation of residual vibration at the design frequency. Experimental results demonstrate RMS residual vibration reductions ranging from 5.3% to 70.7% across seven workspace configurations, with a mean of 43.7% and a maximum peak reduction of 79.1%. A robustness sweep quantifies the shaper's effective frequency range relative to the measured Z-axis natural frequency variation, identifying configurations where a single fixed shaper is sufficient and where configuration-specific tuning would be needed.

2. Methodologies

2.1 System Description

The experimental platform is a PPP Cartesian robot designed for AOI of PCBs. The X and Y axes are belt-driven, each following the signal chain: ESP32 microcontroller → stepper motor driver → NEMA 17 stepper motor (17HS8401) → pulley-belt transmission → linear cart. The Z-axis employs a ball-screw mechanism carrying a camera head assembly of tip mass m_{tip} . A triaxial accelerometer mounted on the camera head records the vibration response during and after positioning moves. The ZVD input shaping algorithm is designed in MATLAB and the resulting impulse sequence is deployed on the microcontroller for real-time command convolution prior to transmission to the stepper motor drivers.

2.2 System Modeling

The belt-driven X and Y axes are modeled as a two-mass system consisting of a motor-side equivalent inertia J_e and a cart of mass M , coupled through a compliant belt

of effective stiffness K_{eff} [11, 12]. Neglecting pulley inertia and damping, the flexible mode resonance frequency of this system is [1,2]:

$$f_{\text{res}} = \frac{1}{2\pi} \sqrt{\frac{K_{\text{eff}}(J_e + MR^2)}{J_e M}} \quad (1)$$

where R is the pulley radius. This resonance is the dominant flexible mode excited by the trapezoidal motion command and is the target of the ZVD shaper design. For shaper design purposes, the belt-drive flexible mode is reduced to a single second-order modal model with the experimentally identified natural frequency ω_n and damping ratio ζ .

The Z-axis arm is modeled as a massless Euler–Bernoulli cantilever of flexural rigidity EI carrying a concentrated tip mass m_{tip} , following the standard lumped-parameter approximation [1,6]. The fundamental natural frequency varies with Z-axis extension z as:

$$\omega_n(z) = \sqrt{\frac{3EI}{m_{\text{tip}} \cdot z^3}} \quad (2)$$

This expression shows that increasing Z extension reduces structural stiffness and shifts the resonant frequency downward, motivating the characterization of f_n across multiple Z positions.

2.3 System Identification

The dominant modal parameters of the X-axis were identified experimentally from post-stop ring-down measurements. Point-to-point moves were executed along the X-axis ($D = 19.2$ cm, $V = 30.0$ cm/s, $A = 220.0$ cm/s²) at three Y-axis positions ($y = 4.5, 9.5$, and 14.5 cm) with the Z-axis at its minimum height position (z_{min} , corresponding to maximum arm extension and minimum structural stiffness). After each move, the accelerometer az signal during the post-stop phase was isolated, the steady-state offset was subtracted, and a 4th-order Butterworth low-pass filter with a cutoff at 35 Hz was applied to isolate the structural vibration mode. The natural frequency f_n was identified from the dominant peak of the FFT of the filtered ring-down signal. The damping ratio ζ was estimated using the logarithmic decrement method [3,6]:

$$\delta = \frac{1}{N} \ln \left(\frac{A_1}{A_{N+1}} \right), \quad \zeta = \frac{\delta}{\sqrt{(2\pi)^2 + \delta^2}} \quad (3)$$

where A_1 and A_{N+1} are the amplitudes of the first and $(N+1)$ -th positive peaks of the ring-down. Three repeated trials were performed at each Y position. The identified parameters, averaged across all nine trials, are $f_n = 15.34 \pm 1.9$ Hz and $\zeta = 0.032 \pm 0.007$. These values served as the design parameters for the ZVD shaper.

2.4 ZVD Shaper Design

Input shaping suppresses residual vibration by convolving the reference command with a sequence of impulses designed to cancel the system's residual oscillation [5]. The Zero Vibration and Derivative (ZVD) shaper constrains both the residual vibration amplitude and its derivative with respect to frequency to zero at the design frequency, yielding greater robustness to modeling errors than the two-impulse ZV shaper [5]. The three-impulse ZVD sequence is given by Singhose [3].

The shaped command is formed as:

$$u_s(t) = \sum_{i=1}^3 A_i \cdot u(t - T_i) \quad (4)$$

where $u(t)$ is the original trapezoidal position command [4]. The shaper adds T_3 to the total move duration, a small penalty relative to the settling time reduction achieved.

A per-Y-position lookup table of ZVD shaper coefficients was implemented on the ESP32 microcontroller, allowing the correct shaper to be selected automatically based on

the current Y-axis position prior to each X-axis move. Design frequencies were determined empirically from preliminary ring-down tests at each Y position. Table 2 summarizes the shaper coefficients deployed on the robot.

Table 1. ZVD shaper parameters deployed per Y-axis configuration with varying X-axis at z_{min} .

Y position (cm)	f_n (Hz)	ζ	A_1	A_2	A_3	$T_2(s)$	$T_3(s)$
14.5	11.1	0.035	0.2782	0.4985	0.2233	0.04507	0.09015
9.5	11.3	0.035	0.2782	0.4985	0.2233	0.04428	0.08855
4.5	11.6	0.035	0.2782	0.4985	0.2233	0.04310	0.08621

Note: f_n and ζ were identified experimentally at each Y position. A_i and T_i correspond to the ZVD shaper coefficients computed for each configuration. z_{min} : Maximum Z-axis extension, minimum structural stiffness.

2.5 Simulation Setup

The flexible mode response was simulated using a second-order base-excitation model driven by the trapezoidal acceleration command $\ddot{u}(t)$ [1,3]. The governing equation during the move is:

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \ddot{u}(t) \quad (5)$$

where x is the relative tip displacement and $\ddot{u}(t)$ is the rigid-body cart acceleration, computed as the second derivative of the trapezoidal position command [3,7]. Initial conditions at the moment of stop, $[x(t_{stop}) \text{ and } \dot{x}(t_{stop})]$, are obtained by integrating Eq. (5) over the move duration using MATLAB's ode45 solver. The post-stop free ring-down from these initial conditions is then computed from the homogeneous form of Eq. (5) and compared between the unshaped and ZVD-shaped cases. The performance metrics are peak residual acceleration $E_{peak} = \max|\ddot{x}(t)|$ and RMS residual acceleration $E_{rms} = \sqrt{\text{mean}(\ddot{x}^2)}$ computed over the post-stop window.

3. Experiments

3.1. Simulations Setup and Procedure

The second-order base-excitation model described in Section 2.5 was simulated in MATLAB R2024b using ode45 numerical integration. For the unshaped and ZVD-shaped commands, the forced ODE was integrated over the trapezoidal move duration ($D = 19.2$ cm, $V = 30.0$ cm/s, $A = 220.0$ cm/s²) to compute the flexible mode displacement and velocity at the moment of stop. The free ring-down from those initial conditions was then simulated and the residual acceleration reconstructed as $\ddot{x}(t) = -\omega_n^2x - 2\zeta\omega_n\dot{x}$. All simulations used $f_n = 11.1$ Hz and $\zeta = 0.035$, matching the shaper deployed on the robot for the z_{min} , $y = 14.5$ cm configuration.

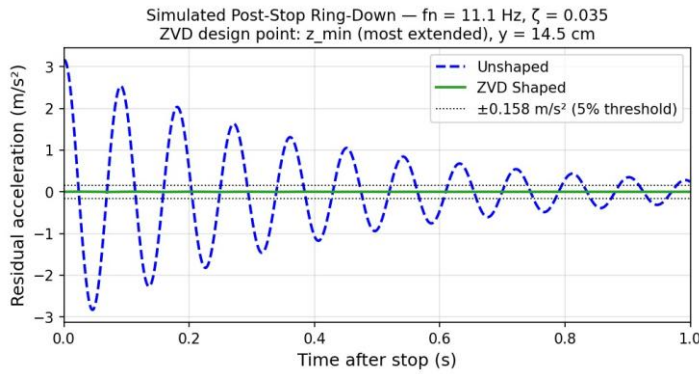


Figure 1. Simulated post-stop ring-down at the ZVD design frequency ($f_n = 11.1$ Hz, $\zeta = 0.035$). The unshaped command leaves significant residual energy at stop, producing a sustained oscillation. The ZVD-shaped command arrives at stop with near-zero flexible mode energy, confirming the cancellation mechanism. Dotted lines indicate the $\pm 5\%$ settling threshold.

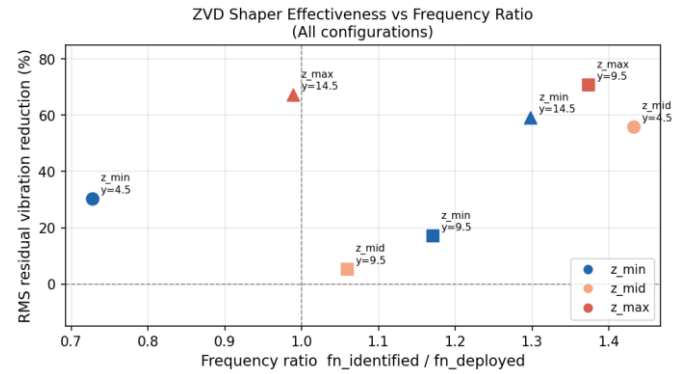


Figure 2. Simulated ZVD scatter plot (RMS residual vibration as percentage of unshaped, normalized to design frequency)

3.2. Experimental Procedure

Point-to-point moves ($D = 19.2$ cm, $V = 30.0$ cm/s, $A = 220.0$ cm/s²) were executed along the X-axis across seven configurations spanning three Z positions (z_{min} , z_{mid} , z_{max}) and three Y positions ($y = 4.5$, 9.5 , and 14.5 cm), with and without the ZVD shaper. Residual vibrations were recorded using an MPU-6050 triaxial accelerometer at 400 Hz via the ESP32 microcontroller. Three repeated trials were performed per configuration where possible. The post-stop accelerometer signal (phase = 2) was low-pass filtered at 35 Hz and the residual metrics E_{peak} and E_{rms} were computed over the first 0.6 s after stop.

Figure 2 shows a scatter plot of RMS reduction versus frequency ratio across all seven configurations. The experimental points broadly follow the predicted trend, with higher reductions occurring where the ratio is near 1.0 or where the baseline vibration amplitude is large enough for the shaper effect to dominate over measurement noise [5,8].

4. Results

4.1. Simulation Results

All simulations used $f_n = 11.1$ Hz and $\zeta = 0.035$, matching the shaper deployed on the robot for the z_{min} , $y = 14.5$ cm configuration. The base-excitation second-order model predicts that the ZVD shaper arrives at the stop moment with near-zero flexible mode energy when the deployed design frequency matches the actual system resonance. Figure 1 shows the simulated post-stop ring-down at the nominal design condition ($f_n = 11.1$ Hz, $\zeta = 0.035$). The unshaped response produces a sustained damped sinusoid decaying over approximately 0.9 s. The ZVD-shaped response begins near zero, confirming theoretical cancellation of residual energy at the design frequency.

The parametric robustness sweep reveals that effective suppression extends over a frequency ratio range of approximately 0.6–1.4, consistent with the known ZVD insensitivity bandwidth [3]. Outside this range, the percentage residual vibration increases rapidly.

4.2. Experimental Results

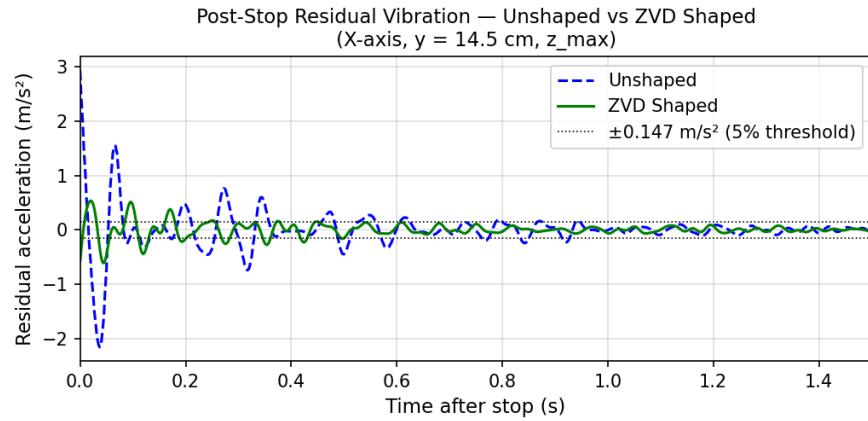


Figure 3. Post-stop residual acceleration at z_{max} , $y = 14.5$ cm — unshaped (dashed) vs. ZVD-shaped (solid). Dotted lines indicate the $\pm 5\%$ settling threshold

Table 3 summarizes the measured residual vibration metrics across all seven validated configurations. RMS residual vibration reductions ranged from 5.3% to 70.7%, with a mean reduction of 43.7% across all configurations. Peak reductions ranged from -4.4% to 79.1% . Settling time at the 5% threshold was reduced by 47.0% at the z_{max} , $y = 14.5$ cm configuration (from 0.950 s to 0.503 s), which exhibited the highest unshaped vibration amplitude in the dataset ($E_{peak} = 2.939$ m/s²).

Figure 3 shows the post-stop ring-down at z_{max} , $y = 14.5$ cm — the configuration with the strongest and most visually clear result. The ZVD-shaped response achieves an immediate and sustained reduction from the first oscillation cycle onward, with E_{peak} reduced from 2.939 m/s² to 0.614 m/s² (79.1%) and E_{rms} reduced from 0.599 m/s² to 0.197 m/s² (67.2%).

The variation in shaper effectiveness across configurations is consistent with the robustness analysis in Section 4.1. Configurations with frequency ratio $\frac{f_{n,identified}}{f_{n,deployed}}$ between 0.73 and 1.43 — that is, all tested configurations — achieved meaningful RMS reductions in five of seven cases. The two cases with weak results (z_{min} , $y = 9.5$ cm: 17.0% RMS; z_{mid} , $y = 9.5$ cm: 5.3% RMS) share a common characteristic: low baseline vibration amplitude relative to the other configurations, which reduces the signal-to-noise ratio of the MPU-6050 measurement and inflates trial-to-trial variability relative to the mean reduction. At z_{min} , $y = 9.5$ cm, the frequency ratio of 1.17 also places this configuration closer to the edge of the robustness band than the other high-performing cases.

Table 1. Experimental residual vibration reduction - all validated configurations.

Z pos.	Y (cm)	f_n ident. (Hz)	f_n dep. (Hz)	Ratio	E_{peak} unshaped (m/s ²)	E_{peak} shaped (m/s ²)	Peak red.	E_{rms} unshaped (m/s ²)	E_{rms} shaped (m/s ²)	RMS red.
z_{min}	4.5	8.44	11.6	0.73	0.515	0.265	48.5%	0.161	0.112	30.4%
z_{min}	9.5	13.23	11.3	1.17	0.399	0.416	-4.4%	0.170	0.141	17.0%
z_{min}	14.5	14.41	11.1	1.30	0.906	0.344	62.1%	0.296	0.121	59.1%
z_{mid}	4.5	16.61	11.6	1.43	0.817	0.411	49.8%	0.289	0.127	56.0%
z_{mid}	9.5	11.97	11.3	1.06	0.793	0.775	2.4%	0.286	0.270	5.3%
z_{max}	9.5	15.52	11.3	1.37	0.871	0.196	77.5%	0.250	0.073	70.7%
z_{max}	14.5	10.98	11.1	0.99	2.939	0.614	79.1%	0.599	0.197	67.2%

Note: $f_{n,ident}$ is the identified natural frequency and $f_{n,dep}$ the deployed one. Peak and RMS values correspond to residual accelerations for unshaped and shaped motions.

The strongest results were obtained at z_{min} , consistent with the Z-axis cantilever model prediction that lower Z positions (longer arm extension) exhibit lower structural stiffness and therefore larger inertial excitation during the aggressive trapezoidal move. The deployed shaper design frequencies (11.1–11.6 Hz) were identified empirically through preliminary ring-down tests, and the subsequent rigorous FFT-based identification revealed actual resonance frequencies of 8.44–16.61 Hz across the tested workspace. Despite this identification discrepancy, the ZVD shaper's inherent robustness to frequency uncertainty [5] ensured effective suppression across most of the tested configurations.

Two configurations were excluded from quantitative analysis: z_{max} , $y = 4.5$ cm, where inspection revealed that the shaped and unshaped data files were identical; and z_{mid} , $y = 14.5$ cm, where no unshaped baseline was available for comparison.

5. Discussion and Conclusions

The results demonstrate that ZVD input shaping is an effective feedforward strategy for residual vibration suppression in a PPP Cartesian robot operating across a variable workspace. A per-Y-position lookup table of ZVD shapers, each designed from empirically identified modal parameters, was implemented on the ESP32 microcontroller and validated across seven configurations spanning three Z positions and three Y positions. Mean RMS residual vibration reduction of 43.7% was achieved across all validated configurations, with individual reductions reaching 70.7% (z_{max} , $y = 9.5$ cm) and 67.2% (z_{max} , $y = 14.5$ cm). Settling time at the 5% threshold was reduced by 47.0% at the highest-amplitude configuration, directly reducing the dead time between positioning moves and image acquisition in the AOI cycle.

The variation in shaper effectiveness across configurations — ranging from near-zero improvement to 71% RMS reduction — is explained by two factors acting simultaneously. First, configurations with higher unshaped vibration amplitude yield larger absolute improvements and clearer signal-to-noise ratios for the MPU-6050 accelerometer, making the shaper effect unambiguous. Second, the frequency ratio between the actual system resonance and the deployed design frequency governs whether the configuration falls within the ZVD robustness band. The experimental scatter plot in Figure 2 confirms that the robustness band predicted by simulation (ratio ≈ 0.6 –1.4) broadly captures the boundary between effective and ineffective suppression in the physical system, validating the second-order base-excitation model as a sufficient approximation for shaper design purposes [5].

A discrepancy exists between the shaper design frequencies (11.1–11.6 Hz) identified through preliminary robot tests and the rigorous FFT-based ring-down identification (8.44–16.61 Hz across the workspace). This discrepancy reflects the difficulty of identifying modal parameters from a noisy low-cost accelerometer under open-loop stepper motor excitation, and highlights the sensitivity of the identification result to the choice of method and signal processing parameters. Despite this discrepancy, the ZVD shaper's inherent robustness to frequency uncertainty ensured effective suppression in most configurations, demonstrating that the method is practical even when the identification is imprecise [3,8].

Several limitations of the present work should be acknowledged. The open-loop stepper motor architecture provides no position feedback; step losses at high accelerations would introduce positioning errors not captured by the model or detectable by the accelerometer. The MPU-6050 accelerometer, while adequate for detecting vibration at configurations with high residual amplitude, had insufficient resolution to produce reliable quantitative results at configurations with low baseline vibration (z_{min} , $y = 9.5$ cm, z_{mid} , $y = 9.5$ cm). Two configurations could not be included in the quantitative analysis

due to data collection issues. The Z-axis cantilever model was not experimentally validated due to sensor resolution constraints at the low vibration amplitudes present in Z-axis motion.

Future work should address these limitations through three directions. First, replacing the MPU-6050 with a higher-resolution inertial measurement unit or a laser displacement sensor would enable reliable identification and validation across the full workspace, including low-amplitude configurations. Second, an adaptive shaping strategy that updates shaper parameters in real time from ring-down measurements after each move would eliminate the reliance on preliminary identification and handle workspace-dependent frequency shifts continuously [3]. Third, extending the experimental validation to cover the full AOI scan trajectory — including combined X and Y moves — would demonstrate the method's applicability to the complete inspection cycle.

In conclusion, this work presents the design, simulation, and experimental validation of a ZVD input shaper for a PPP Cartesian AOI robot with variable Z-axis dynamics. The shaper was implemented on a low-cost ESP32 microcontroller with per-configuration lookup table selection, requiring no additional sensors beyond the onboard accelerometer. Experimental results across seven workspace configurations confirm residual vibration reductions of up to 79.1% in peak and 70.7% in RMS, demonstrating the practical feasibility of input shaping for throughput improvement in PCB inspection systems.

References

1. Mehmood, Y.; Cannella, F.; Cocuzza, S. Analytical Modeling, Virtual Prototyping, and Performance Optimization of Cartesian Robots: A Comprehensive Review. *Robotics* 2025, 14, 62. <https://doi.org/10.3390/robotics14050062>
2. Jokinen, M. Centralized Motion Control of a Linear Tooth Belt Drive: Analysis of the Performance and Limitations. Ph.D. Dissertation, Acta Universitatis Lappeenrantaensis 407, Lappeenranta University of Technology, Lappeenranta, Finland, 26 November 2010; ISBN 978-952-265-000-9. <https://lutpub.lut.fi/handle/10024/66177>
3. Singhose, W. Command Shaping for Flexible Systems: A Review of the First 50 Years. *Int. J. Precis. Eng. Manuf.* 2009, 10, 153–168. <https://doi.org/10.1007/s12541-009-0084-2>
4. Singer, N.C.; Seering, W.P. Preshaping Command Inputs to Reduce System Vibration. *J. Dyn. Syst. Meas. Control* 1990, 112, 76–82. <https://doi.org/10.1115/1.2894142>
5. Meckl, P.H.; Seering, W.P. Experimental Evaluation of Shaped Inputs to Reduce Vibration for a Cartesian Robot. *J. Dyn. Syst. Meas. Control* 1990, 112, 159–165. <https://doi.org/10.1115/1.2896122>
6. Rao, S.S. *Mechanical Vibrations*, 6th ed.; Pearson: Hoboken, NJ, USA, 2018; pp. 317–320. ISBN 978-0-13-468169-0.
7. Singhose, W.E.; Derezinski, S.J.; Singer, N.C. Extra-Insensitive Input Shapers for Controlling Flexible Spacecraft. *J. Guid. Control Dyn.* 1996, 19, 385–391. <https://doi.org/10.2514/3.21630>
8. Singhose, W.; Seering, W.; Singer, N. Input Shaping for Vibration Reduction with Specified Insensitivity to Modeling Errors. In *Proceedings of the 1996 Japan–USA Symposium on Flexible Automation*, Boston, MA, USA, 7–10 July 1996; ASME: New York, NY, USA; Vol. 1, pp. 307–313.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.